

THE Intelligence Analyst's Playbook

sans.org/osint

Poster Created by Matt Edmondson. ©2026 SANS Institute. All Rights Reserved.
CD_OSINT_02-26

Intelligence Product Structure – BLUF

1

Bottom Line Up Front (BLUF)
Make your principal judgment in 2-3 sentences. State what you assess, at what confidence level, and the single most important implication. This is the paragraph the decision maker may read and nothing else.

2

Key Judgments
Make 3-5 numbered analytical judgments, each with a stated confidence level. These are your assessed conclusions, not facts. Each should be independently defensible.

3

Evidence Summary
What do we know and how do we know it? Summarize the evidence base, source types, and quality. Identify the strongest and weakest links in your evidence chain.

4

Analysis
This is the detailed reasoning that connects evidence to judgments. Show your analytical work—which SATs were applied, how evidence was weighed, and what patterns were identified. Evidence-to-judgment traceability must be explicit.

5

Alternatives
What competing hypotheses did you consider and why were they assessed as less likely? What evidence would change your mind? This section demonstrates analytical rigor and intellectual honesty.

6

Implications
So what? What does your assessment mean for the decision maker?—operational, strategic political, and resource implications. Recommend course of action if requested.

7

Outlook
What happens next? Give key indicators and signposts to monitor, alert trigger points for reassessment and the timeframe for expected developments.

If the decision maker reads only the first paragraph, they should have your assessment.

Confidence Language

Almost certain . . . 93–99%
Very likely 85–92%
Likely 70–84%
Probable 55–69%

Roughly even 45–54%
Unlikely 15–44%
Very unlikely 3–14%
Remote 1–2%

Source Attribution Format

Use this format for in-text source attribution:
According to [Source LUPUL] AI – Completely Reliable/Confirmed), PETRESCU is the operational leader of the VULPE network [REF: LUPUL-202-0047, dated DD/MM/YY].

Always include:
• Source designation • Reference number
• Admiralty rating • Date of information

Writing Checklist

■ Does BLUF state the assessment clearly?
■ Does every judgment have a confidence level?
■ Does every judgment trace back to evidence?
■ Are alternatives addressed?
■ Is active voice used throughout?
■ Are there no unsupported assertions?
■ Are implications stated?
■ Are the monitoring indicators defined?

THE BLUF DISCIPLINE

Intelligence products are not academic papers. Put your conclusion first, your evidence second. Decision makers need your assessed judgment immediately. They will read the supporting evidence only if they choose to—structure accordingly.

The CRAAP Test – Source Evaluation Framework

Five criteria for evaluating the reliability and value of any information source

WHAT IS THE CRAAP TEST?

A structured evaluation framework developed at California State University—Chico for assessing information quality. In intelligence analysis, it provides a repeatable, auditable method for grading sources before they enter the analytical process. Every piece of evidence in your ACH matrix, every data point in your timeline, every claim in a report should survive CRAP testing before you trust it. The framework: evaluates five dimensions: **Currency**, **Relevance**, **Authority**, **Accuracy**, and **Purpose**.

Quick Scoring Guide

45–50 **Excellent**

35–44 **Good**

25–34 **Acceptable**

15–24 **Marginal**

5–14 **Reject**

Score each criteria 1–10 with a total cut of 50. Always document your reasoning, not just the number.

Currency

EVALUATE
• When was the information published or last updated?
• Has the situation changed since publication?
• Are the links and references still functional?
• Does the topic require current or historical information?

INTELLIGENCE CONTEXT

• A 2019 assessment of hybrid threat tactics may be outdated by 2025
• Financial data older than 90 days may miss critical shifts
• Satellite imagery degrades in value hourly during a crisis

RED FLAGS:

• No publication date • Links are broken • References cite sources 5+ years old with no updates • Timeless framing on a rapidly evolving topic

Relevance

EVALUATE
• Does it directly address your intelligence question?
• Who is the intended audience?
• Is the information at the right level of detail?
• Does it add something your other sources don't?
• Would you be comfortable citing it in a product

INTELLIGENCE CONTEXT

• A source about general crime trends may not speak to your specific network
• Academic research may be too theoretical for operational needs
• Tactical reporting may miss the strategic picture

RED FLAGS:

• Tangential to your actual question • Written for a completely different audience • Too general to be actionable • Duplicates what you already have

Authority

EVALUATE
• Who is the author and what are their credentials?
• What organization published this?
• Is there contact information or institutional affiliation?
• Does the author have recognized expertise in this area?
• Has this source been reliable in the past?

INTELLIGENCE CONTEXT

• An anonymous telegram channel has near-zero authority
• A ministry report carries institutional weight but may have political bias
• Track record matters—has this source been wrong before?

RED FLAGS:

• Anonymous author • No institutional backing • Self-published with no peer review • Domain doesn't match claimed authority (e.g., .com for government data)

Accuracy

EVALUATE
• Is the information supported by evidence?
• Can claims be verified through other sources?
• Has it been reviewed or vetted by others?
• Are there factual errors, inconsistencies, or typos?
• Does the tone seem measured or emotionally charged?

INTELLIGENCE CONTEXT

• Single-source reporting is inherently less accurate than corroborated intelligence
• Statistics without methodology should be treated with suspicion
• Corroboration from independent sources increases confidence

RED FLAGS:

• No citations or sources given • Claims are extraordinary but evidence is thin • Numbers don't add up • Contradicts well-established reporting

Purpose

EVALUATE
• What is the intent—inform, persuade, sell, entertain, or deceive?
• Is the point objective or biased?
• Are conflicts of interest disclosed?
• Does the language reveal an agenda?
• Who benefits from you believing this?

INTELLIGENCE CONTEXT

• State media exists to advance state narratives
• Leaked documents may be genuine but selectively released to shape perception
• Disinformation is designed to pass the first four tests and fail only this one

RED FLAGS:

• Emotionally manipulative • One-sided presentation • Source has financial or political stake in the conclusion • Conveniently confirms what the adversary wants you to believe

The Intelligence Cycle

Six phases of the continuous intelligence process—from requirement to decision and back again

The cycle is not a conveyor belt. Any phase can trigger a return to any earlier phase. New evidence during **Analysis** may send you back to **Collection**. A decision-maker's feedback after **Dissemination** may redefine **Direction** entirely.

The intelligence cycle is not six steps you complete—it is a **continuous conversation** between those who need to know and those who find out.



- Where Cycles Break Down**
- Direction phase skipped—analysts collect without clear requirements
 - A Collection drives analysis instead of requirements driving collection
 - Processing bottleneck—raw data sits untranslated or uncatalogued
 - Analysis skipped—raw reports forwarded as “intelligence”
 - Feedback loop missing—products delivered but never evaluated
 - Stovepiping—phases run in isolation without cross-phase communication

Direction

“What do we need to know?”

The decision maker defines intelligence priorities. Without clear direction, the entire cycle produces noise instead of insight.

KEY ACTIVITIES
• Define key intelligence questions (KIOs)
• Establish priority intelligence requirements
• Set timelines and output formats
• Identify stakeholder and consumers

ANALYST'S ROLE
• Translate vague requests into answerable questions
• Push back on questions that are too broad or unfocused
• Negotiate realistic scope and deadlines
• Draft a mission statement for the tasking

ITERATIVE, NOT LINEAR
The cycle is not a production line. Every phase can loop back to any earlier phase. New evidence in Phase 4 may require new collection in Phase 2.

Collection

“Where do we find it?”

Systematic gathering of information from all available sources, guided by the collection plan derived from **Direction**.

SOURCE TYPES
• **HUMINT**—Human sources, informants, interviews
• **SIGINT**—Signals, intercepts, metadata
• **OSINT**—Open sources, media, social media
• **FININT**—Financial records, transaction flows
• **IMINT**—Imagery, satellite, surveillance

PRINCIPLES
• Collect against requirements, not curiosity
• Use multiple independent sources
• Document source reliability (Admiralty Code)
• Identify collection gaps early
• The collection plan is a living document

Processing

“How do we make it usable?”

The decision maker defines intelligence priorities. Without clear direction, the entire cycle produces noise instead of insight.

KEY ACTIVITIES
• Translation and transcription
• Decryption and technical exploitation
• Data reduction and filtering
• Indexing, cataloging, and cross-referencing

CRITICAL POINTS
• Unprocessed data is not intelligence
• Processing delays can make time-sensitive collection worthless
• Quality control here prevents garbage-in analysis
• Metadata is often as valuable as content

REQUIREMENTS-DRIVEN
Collection without direction is noise. Analysis without requirements is academic exercise. Everything starts with a clear question from a decision maker.

Analysis

“What does it mean?”

This is the intellectual core of the cycle. Processed information is evaluated, integrated, and interpreted to produce assessed judgments.

STRUCTURED TECHNIQUES
• ACH—competing hypothesis
• Key assumptions check
• Timeline/link analysis
• Process mapping and decomposition
• Red teaming and devil's advocacy

ANALYST DISCIPLINE
• Separate what you know from what you think
• State confidence levels explicitly
• Consider alternative explanations
• Guard against cognitive biases
• Document your reasoning chain

CONSUMER-FOCUSED
The analyst serves the decision maker, not the other way around. Products must be timely, relevant, actionable, and at the right classification level.

Production

“How do we present it?”

Analytical conclusions are packaged into a structured intelligence product designed for the specific consumer.

PRODUCT ELEMENTS
• Bottom line up front (BLUF)
• Key judgments with confidence levels
• Evidence and analytical reasoning
• Alternative scenarios and indicators
• Recommendations (if appropriate)

QUALITY STANDARDS
• Calibrated probability language throughout
• Source transparency—show your evidence
• Appropriate classification and handling
• Tailored to audience—minister versus operator
• Peer review before release

FEEDBACK IS MANDATORY
A cycle without feedback is an assembly line. Decision makers must tell analysts what worked, what didn't, and what they need next. Demand it.

Dissemination

“Did it reach the right person?”

Delivering the finished product to the right consumer, in the right format, at the right time. A brilliant product delivered late is a failed product.

DELIVERY CONSIDERATIONS
• Format: written brief, oral briefing, dashboard
• Classification and need-to-know restrictions
• Timeliness—is it still actionable?
• Distribution list—who else needs this?

CLOSING THE LOOP
• Solicit feedback: was it useful? Actionable?
• Did it answer the original requirement?
• What new questions did it generate?
• Feed lessons back into Direction for next cycle

Calibrated Probability Language

Use standardized language to communicate likelihood. Every probability term maps to a defined numerical range. When writing intelligence products, always use these terms consistently. The reader must understand your confidence level without ambiguity.

ALMOST CERTAIN	93%–99%
VERY LIKELY	85%–92%
LIKELY	70%–84%
PROBABLE	55%–69%
ROUGHLY EVEN	45%–54%
UNLIKELY	15%–44%
VERY UNLIKELY	3%–14%
REMOTE	1%–2%

Correct Usage
“We assess it as **LIKELY** (70%–84%) that the VULPE network is expanding its operations into Arad county.”
“It is **VERY UNLIKELY** (3%–14%) that the disinformation campaign is entirely organic.”
Avoid These
“We believe the network might possibly be expanding.”
“It is certain that MCG is state-directed.”
“There is no chance the route will remain active.”

Tip: When your assessment falls at a boundary between terms, use the lower confidence term and explain why. Understating confidence is always preferable to overstating it. Include the numerical range in parentheses after the term when precision matters.

Admiralty/NATO Source and Information Evaluation

Every piece of intelligence must be evaluated on two independent axes: the *reliability of the source* and the *credibility of the information*.

		INFORMATION CREDIBILITY					
		1 Confirmed	2 Probably True	3 Possibly True	4 Doubtful	5 Improbable	6 Cannot Be Judged
SOURCE RELIABILITY	A Completely Reliable	A1 Highest Confidence	A2 Very High	A3 High	A4 Moderate	A5 Low	A6 Cannot Assess
	B Usually Reliable	B1 Very High	B2 High	B3 Moderate–High	B4 Moderate	B5 Low	B6 Cannot Assess
	C Fairly Reliable	C1 High	C2 Moderate–High	C3 Moderate	C4 Moderate–Low	C5 Low	C6 Cannot Assess
	D Not Usually Reliable	D1 High	D2 Moderate–High	D3 Moderate	D4 Moderate–Low	D5 Low	D6 Cannot Assess
	E Unreliable	E1 Low	E2 Low	E3 Very Low	E4 Very Low	E5 Minimal	E6 Cannot Assess
	F Cannot Be Judged	F1 Cannot Assess	F2 Cannot Assess	F3 Cannot Assess	F4 Cannot Assess	F5 Cannot Assess	F6 Cannot Assess
		CONFIDENCE LEVEL: High High–Moderate Moderate Moderate–Low Low Cannot Assess					

Source Reliability (A–F)
Assessed based on the source's *track record*, *access*, *motivation*, and *past accuracy*. Ask: Has this source been tested before? Do they have direct access to the information? What is their motivation for sharing?

Information Credibility (1–6)
Assessed based on the information's *internal consistency*, *corroboration*, *plausibility*, and *timeliness*. Ask: Does it align with other reporting? Is it internally consistent? Is it plausible given what we know?

KEY PRINCIPLE:
Rate the **SOURCE** and the **INFORMATION** independently. A reliable source can provide inaccurate information. An unreliable source can provide accurate information.

Analysis of Competing Hypotheses (ACH)

A structured technique for evaluating multiple explanations against evidence

WHAT IS ACH?

A systemic method that forces analysts to consider all plausible hypotheses simultaneously, evaluate evidence against each, and identify which hypotheses are most inconsistent with the evidence—rather than seeking to confirm a preferred explanation. ACH combats confirmation bias by focusing on disconfirming evidence—the most diagnostic kind.

Developed by Richards J. Heuer Jr, CIA (1999)

The Eight-Step Process

1 Identify Hypotheses

Brainstorm all plausible explanations—minimum of three and ideally more than five. And include unlikely ones too.

2 List Evidence and Arguments

Gather all relevant evidence, assumptions, and logical arguments—include absence of expected evidence.

3 Build the Matrix

Have the hypotheses across columns and evidence down rows. Assess each cell independently.

4 Score Consistency

Mark each cell: Consistent (C), Inconsistent (I), or Not Applicable (N). Work across rows and not down columns.

5 Refine the Matrix

Delete evidence consistent with all hypotheses—it has no diagnostic value. Focus on discriminating evidence.

6 Draw Tentative Conclusions

The hypothesis with the least inconsistency is the leading candidate. Reject those with most “I”s.

7 Assess Sensitivity

How sensitive is the conclusion? If one piece of evidence changed, would the answer flip? Flag fragile judgments.

8 Report and Monitor

Present conclusions with confidence levels. Identify milestones or indicators that would change the assessment.

Core Principles

1

Disprove, don't prove. The most diagnostic evidence is that which eliminates hypotheses, not that which confirms your favorite.

2

Work across rows. Evaluate each piece of evidence against ALL hypotheses before moving to the next. Never work down a single column.

3

Absence matters. Evidence you expect to see but don't is just as important as evidence you do see. Include it in the matrix.

4

Reject the rejected. A hypothesis with even one clear inconsistency is weakened. Multiple inconsistencies should eliminate it entirely.

5

Test sensitivity. If removing one piece of evidence changes your conclusion, that judgment is fragile—say so.

Common Pitfalls

✗

Starting with only two hypotheses—it forces binary thinking

✗

Working down columns instead across rows

✗

Including only evidence that supports the lead hypothesis

✗

Treating “consistent” as “proves”—consistency is weak

✗

Ignoring absent evidence or evidence of absence

✗

Leaving rejected hypotheses in without formally eliminating them

When to Use ACH

✓

Multiple plausible explanations exist for observed events

✓

Deception or denial may be involved

✓

Analysts disagree on the most likely explanation

✓

High-stakes decisions depend on the assessment

✓

You need an auditable, transparent reasoning trail

		INDICATOR				
		H1 State-Directed Hybrid Operation	H2 Organized Crime Independent	H3 Crime-State Nexus	H4 Internal Corruption Only	
EVIDENCE	SIGINT	Encrypted communications between Varga and foreign contact	C	I	C	I
	FININT	Shell company payments EUR 180K–620K/month	C	C	C	N
	HUMINT	Source reports Varga met foreign intelligence officer	C	I	C	I
	OSINT	Coordinated disinformation campaign timing matches election cycle	C	N	C	N
	SIGINT	No direct link between Varga and state ministry officials	I	C	N	C
	HUMINT	Varga's network uses known OCG smuggling routes	N	C	C	N
	FININT	Payments routed through jurisdictions linked to state proxies	C	I	C	I
	OSINT	Judge Albescu's rulings favor Varga-linked entities	N	C	C	C
ASSESSMENT		WEAKENED 1 inconsistency	REJECTED 3 inconsistencies	STRONGEST 0 inconsistencies	REJECTED 3 inconsistencies	
KEY: C = Consistent I = Inconsistent N = Not Applicable		SOURCES: SIGINT HUMINT OSINT FININT				

READING THE MATRIX

H3 (Crime-State Nexus) has zero inconsistencies and is consistent with all evidence. H1 (State-Directed) remains plausible with one inconsistency; H2 and H4 are rejected—three pieces of evidence directly contradict each. The key diagnostic evidence is the foreign intelligence link (eliminates H2 and H4) and the absence of direct state-ministry ties (weakens H1).

Conclusion: The Vulpes network most likely operates as a crime-state nexus—criminal infrastructure serving foreign state interests through deniable intermediaries.

STEEP+S Scanning Framework

Seven sectors for systematic horizon scanning—ensuring no blind spots in strategic analysis

Scan ALL sectors systematically. The threat you miss is the one outside your usual lens.

1

Security

Threats to state and citizen safety: military, terrorism, organized crime, border security, cyber threats

SCANNING QUESTIONS

- Are new armed groups or alliances forming?
- Are border control capacities changing?
- Are crime-state nexus patterns evolving?

EXAMPLE INDICATORS

- Armed confrontation frequency at borders
- Intelligence cooperation agreement changes
- Foreign intelligence officer activity levels

5

Political

Government stability, elections, EU/NATO alignment, foreign policy shifts, and political influence operations

SCANNING QUESTIONS

- Are election timelines creating vulnerability windows?
- Are EU accession processes stalling or accelerating?
- Are proxy political movements emerging?

EXAMPLE INDICATORS

- Polling data on EU/NATO support
- New political movement follower growth rates
- Government of neighbouring states changing

2

Technology

Emerging technologies that alter the threat landscape: cyber capabilities, AI, surveillance, communications

SCANNING QUESTIONS

- Are adversaries acquiring new technical capabilities?
- Are deepfake or AI tools being deployed?
- Are encrypted communication patterns shifting?

EXAMPLE INDICATORS

- Deepfake production capability acquisition
- Shift from text to video disinformation
- New encrypted channel adoption rates

6

Legal

Legal frameworks, judicial independence, international law, extradition, and regulatory environments

SCANNING QUESTIONS

- Are judicial figures being co-opted or compromised?
- Are legal frameworks for countering hybrid threats adequate?
- Are international cooperation mechanisms functioning?

EXAMPLE INDICATORS

- Judicial contact with foreign influence entities
- Extradition cooperation response times
- Legislative changes affecting intelligence mandates

3

Economic

Financial flows, economic dependencies, trade, sanctions, and illicit finance that shape the strategic environment

SCANNING QUESTIONS

- Are unusual financial flows appearing through shell companies?
- Are energy dependencies creating leverage points?
- Is economic deterioration fueling grievances?

EXAMPLE INDICATORS

- Financial flow volume changes (e.g., EUR 180K to 620K/month)
- New corporate registrations in offshore jurisdictions
- Energy price and supply dependency metrics

7

Societal

Public opinion, social cohesion, demographics, media trust, radicalization, and civil society health

SCANNING QUESTIONS

- Are anti-institutional narratives gaining traction?
- Are vulnerable communities being targeted for radicalization?
- Is trust in media and institutions declining?

EXAMPLE INDICATORS

- Anti-EU content share of social media (e.g., 58%)
- Telegram channel subscriber growth rates
- Petition and protest participation trends

4

Environmental

Climate, geography, resources, and environmental changes affecting security and migration patterns

SCANNING QUESTIONS

- Are climate changes altering migration corridors?
- Are resource scarcities creating new conflict drivers?
- Are natural disasters opening exploitation opportunities?

EXAMPLE INDICATORS

- Migration route shifts due to seasonal changes
- Resource competition in transit corridor regions
- Maritime incident frequency in Black Sea

OSINT Community <http://sansurl.com/osint>

Resources sans.org/osint

Videos youtube.com/@SANScyberDefense

Blogs sans.org/blog

Webcasts sans.org/webcasts



SANS OSINT Training Courses



Dator's Four Futures

Four archetypes of the future—a completeness framework for scenario development

APPLICATION:

Use these four archetypes as a completeness check. When developing scenarios for strategic assessment, ask: “Have we considered all four types? If your scenario set only contains variations on continuation, you are missing the futures that matter most for resilience planning. Every scenario falls into one of these four archetypes:

Continuation “What do we need to know?” Current trends persist and extrapolate forward. Existing power structures, growth patterns, and trajectories continue without fundamental disruption. Change occurs incrementally within existing systems. Most forecasts implicitly assume continuation—this is both its strength (often correct) and its danger (blindspot for discontinuity).	Collapse “Things fall apart” Systems break down or fail catastrophically. This may be economic collapse, institutional failure, environmental crisis, or security breakdown. Collapse scenarios test what happens when the assumptions underlying “continuation” prove fragile. The failure may be sudden or a cascading sequence of interconnected failures across domains.	Discipline “Someone takes control” Order is imposed by a strong state, an ideology, a regulatory framework, or an external power. Resources are rationed, freedoms constrained, and systems managed centrally. Discipline futures emerge in response to perceived chaos or crisis. They offer stability at the cost of autonomy. Authoritarian responses to hybrid threats fail here.	Transformation “New paradigm emerges” Fundamental change creates entirely new systems, values, or ways of organizing society. This is not a return to an earlier state but a genuine shift: technological, social, or political. Transformation may be positive (democratic renewal, new security architecture) or disorienting (societal upheaval). The key feature is discontinuity from the present.
---	---	--	--

Cognitive Biases in Intelligence Analysis

Systematic errors in thinking that distort analytical judgment—and how to fight them

WHY BIASES MATTER IN INTELLIGENCE

Cognitive biases are not character flaws—they are built-in features of human cognition that normally help us make fast decisions under uncertainty. In intelligence analysis, these same shortcuts become dangerous because they cause analysts to see patterns that aren't there, dismiss evidence that contradicts expectations, and anchor on first impressions. Biases are invisible to the person experiencing them. You cannot think your way out of a bias you don't know you have. The only reliable defense is structured process—techniques that force consideration of alternatives, disconfirming evidence, and external perspectives.

Kahneman's Two Systems

System 1 – Fast

Automatic, intuitive, effortless. Pattern recognition, gut feelings, snap judgments. Operates below conscious awareness. Source of most biases.

System 2 – Slow

Deliberate, analytical, effortful. Logical reasoning calculation, critical evaluation. Requires conscious attention. Lazy—defaults to System 1 when possible.

Most analytical errors occur when System 1 answers a question that should have been given to System 2.

Search and Attention How we look for information	Evaluation and Estimation How we weigh evidence	
CONFIRMATION BIAS Seeking, noticing, and remembering evidence that supports your existing belief while ignoring what contradicts it <i>"I knew Varga was state-linked"—then only collecting evidence of state ties while dismissing CCG indicators</i>	MIRROR IMAGING Assuming the adversary thinks, values, and decides the way you do. Projecting your own rationality onto others. <i>"They wouldn't risk that—the cost is too high" but the adversary may not have shared your cost calculus</i>	
ANCHORING Over-relying on the first piece of information encountered, which then frames all subsequent analysis <i>The initial source report says "likely Russian"—every later assessment is unconsciously measured against that anchor</i>	HINDSIGHT BIAS After an event, believing you "knew it all along." Rewriting your memory of prior uncertainty. <i>"The warning signs were obvious"—but only because you now know the outcome. Before the event, the signals were buried in noise.</i>	
AVAILABILITY BIAS Judging the probability of events by how easily examples come to mind, not by actual frequency <i>A recent high-profile cyber attack makes the analyst overweight cyber threats and underweight physical infiltration</i>	OVERCONFIDENCE Excessive certainty in your own judgments. Assigning narrower probability ranges than the evidence supports. <i>Saying "almost certain" when the evidence only supports "likely"—because you trust your intuition more than the data</i>	
ATTENTION BIAS Focusing on dramatic or emotionally vivid information while overlooking routine data that may be more diagnostic <i>A leaked video consumes all analytical attention while a pattern of quiet financial transfers goes unnoticed</i>	PROPORTIONALITY BIAS Assuming big events must have big causes. Small actors cannot produce large consequences. <i>Discussing the idea that a single corrupt judge could unravel an entire border security framework</i>	
Social and Group How teams distort judgment	Output and Action How we communicate and decide	
GROUPTHINK The desire for team harmony suppresses dissent, alternative views, and critical evaluations. The group converges on a comfortable answer. <i>Nobody challenges the cell lead's hypothesis because they don't want to slow the team down before the deadline</i>	SATISFICING Accepting the first "good enough" explanation and stopping the search for better alternatives. <i>The first hypothesis fits the evidence reasonably well so the analyst stops looking for a better explanation</i>	
AUTHORITY BIAS Deferring to the senior person's view regardless of evidence. Rank substitutes for analysis. <i>The experienced analyst says it's state-directed and the junior analyst silently shelves the OCG evidence</i>	SUNK COST FALLACY Continuing to defend a position because you've already invested time and reputations in it, not because evidence supports it. <i>The team spent three days building a timeline around hypothesis A—abandoning it now feels like "wasted work"</i>	
BANDWAGON EFFECT Adopting a view because others hold it. The more people believe something, the more "true" it feels. <i>Three partner agencies report the same conclusion (from the same single source)—and now it feels like consensus</i>	FRAMING EFFECT How information is presented changes the conclusion drawn. The same data framed differently produces different analytical judgments. <i>"15% chance of failure" versus "85% chance of success"—same statistic, different analytical response</i>	
NOT INVENTED HERE Rejecting information analysis or methodology because it comes from outside your team or agency. <i>Dismissing a partner service's assessment because they don't understand our operating environment</i>	STATUS QUO BIAS Preference for the current assessment. Change requires effort and risk, so the default is to maintain the existing positions. <i>The standing assessment says "low threat" and the analyst needs overwhelming evidence to change it, even when the threshold should be lower</i>	
Structured Countermeasures ✓ Analysis of Competing Hypothesis Forces simultaneous evaluation of all explanations. It counts confirmation bias and satisficing. ✓ Devil's Advocacy Assign someone to argue the opposite case as strongly as possible. It counters groupthink and authority bias. ✓ Key Assumptions Check Explicitly list what you're taking for granted. It counters anchoring and mirror imaging. ✓ Pre-Mortem Analysis Imagine your conclusion is wrong—then explain why. It counters overconfidence and hindsight bias.	Common Pitfalls ✗ Layered Confirmation Biased collection feeds biased analysis feeds biased production—each phase reinforcing the error ✗ Consensus as Proof Five analysts who read the same single-source report do not represent five independent confirmations ✗ Absence Blindness Failing to notice what DIDN'T happen—the dog that didn't bark is often the most important evidence ✗ Expertise Trap Experience analysts are MORE susceptible to anchoring and overconfidence, not less ✗ Time Pressure Default Under deadline stress, System 2 shuts down and System 1 runs the analysis unchecked	Structured Countermeasures ? What evidence would change my mind? ? Am I looking for evidence to confirm or to test my hypothesis? ? Would I assess this differently if the first report I read had said the opposite? ? Is my confidence based on evidence quality or evidence volume? ? Have I genuinely considered the strongest version of the alternative view? ? Am I maintaining this position because the evidence supports it or because I've already committed to it? ? If I am wrong, what are the consequences? Is the cost of error worth the current confidence level?
You cannot eliminate biases. You can only structure your process so that biases are surfaced, challenged, and documented before they reach the decision maker.		

SEC497: Practical Open-Source Intelligence (OSINT)

SEC497 is based on two decades of experience with OSINT research and investigations supporting law enforcement, intelligence operations, and a variety of private sector businesses ranging from small start-ups to Fortune 100 companies. The goal is to provide practical, real-world tools and techniques to help individuals perform OSINT research safely and effectively. One of the most dynamic aspects of working with professionals from different industries worldwide is getting to see their problems and working with them to help solve those problems. SEC497 draws on lessons learned over the years in OSINT to help others. The course not only covers critical OSINT tools and techniques, it also provides real-world examples of how they have been used to solve a problem or further an investigation. Hands-on labs based on actual scenarios provide students with the opportunity to practice the skills they learn and understand how those skills can help in their research. SEC497 includes 29 hands-on labs and a capstone capture-the-flag event.

GIAC Certification: GIAC Open Source Intelligence (GOSI)

SEC587: Advanced Open-Source Intelligence (OSINT) Gathering and Analysis

With OSINT being the engine of most major investigations in this digital age, the need for a more advanced course was inevitable. The data in almost every OSINT investigation becomes more complex to collect, exploit, and analyze. To address this reality, OSINT practitioners must be able to perform OSINT at scale and possess the means and methods to check and report on the reliability of their analysis. In SEC587 you will learn how to perform advanced OSINT gathering and analysis as well as understand and use common programming languages such as JSON and Python. SEC587 also will go into dark web and financial (cryptocurrency) topics as well as disinformation, advanced image, and video OSINT analysis. This is an advanced, fast-paced course that will give seasoned OSINT investigators new techniques and methodologies and entry-level OSINT analysts that extra depth in finding, collecting, and analyzing data sources from all around the world.

GIAC Certification: GIAC Strategic OSINT Analyst (GSOA)